Harshit Shah

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PROFILE SUMMARY

AI Engineer with experience of building and shipping production-grade AI systems spanning LLM applications, RAG pipelines and NL2SQL interfaces. Experienced in fine-tuning, prompt engineering, and deploying end-to-end ML solutions that solve real-world problems. Published researcher with a track record of measurable impact across multiple products.

EDUCATION

MIT ADT University, Pune, India Aug 2021 - Jul 2025
B.Tech in Biomedical Engineering

SKILLS

Technical : Machine Learning, LLM Fine-tuning, RAG, NL2SQL, Prompt Engineering, MLOps, Agentic AI
AI / ML : PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, LangChain, LangGraph, OpenCV, FastAPI
Languages / Tools : Python, SQL, Java, TypeScript, React, Pinecone, Vector Databases, Docker, AWS, Git/GitHub, CrewAI

PROFESSIONAL EXPERIENCE

Actorius Innovations and Research – Pune Oct 2025 – Mar 2026
AI Engineer

- Engineered a production-deployed WhatsApp AI assistant using LLM + RAG, supporting multi-turn FAQ resolution, automated test booking, and intelligent live-agent escalation for real users at production scale.
- Built a desktop CTC analysis application that processes .czi (Carl Zeiss) microscopy files through a marker-first pipeline, automating fluorescence channel parsing, cell segmentation, and multi-class classification.
- Designed an NL2SQL system using a fine-tuned transformer with schema-aware prompting, enabling the clinical team to query structured patient datasets in plain English without any SQL knowledge.

1Cell.Ai - Pune Jul 2025 - Oct 2025
Machine Learning Engineer

- Built a document intelligence pipeline processing 500+ oncology PDFs/month using PyPDF2 and pdfplumber with 12+ regex patterns for genomic entity extraction (SNVs, CNVs, fusions, VAF, MSI, TMB, HRD), achieving 95% accuracy and 3x throughput via parallel processing.

1Cell.Ai - Pune Jan 2025 - Jun 2025
Project Intern

- Automated CTC image analysis using ImageJ macros, Java plugins, and Bio-Formats integration, achieving a 140x performance improvement from 7 minutes to 3 seconds per sample; published findings in ESMO Open (2025).

Bioinformatics Centre, SPPU - Pune Jul 2024 - Sep 2024
Research Intern

- Performed DESeq2 differential expression analysis on 522 TCGA RNA-seq samples, identified 117 DEGs, and surfaced 10 hub gene biomarker candidates via Cytoscape PPI network analysis, validated across 9,736 samples on GEPIA.

PROJECTS

TrendSense — Multi-Agent Market Intelligence Platform

- Architected a multi-agent market intelligence platform using LangGraph StateGraph, orchestrating a parallelized Fetcher → Sentiment → Synthesis → RAG-Validation pipeline that concurrently ingests 10,000+ signals from Reddit, HackerNews, and NewsAPI via Python asyncio.
- Built a self-correcting RAG agent backed by ChromaDB that fact-checks LLM-generated trend summaries against historical vector embeddings, preventing hallucinations and filtering already-mainstream signals — powered by Groq (Llama 3.3) for low-latency inference.
- Developed a custom S-TVS (Sentiment-Adjusted Trend Velocity Score) algorithm using a dual-model sentiment engine (VADER + TextBlob); served results via FastAPI SSE-streaming backend with a React 18 + TypeScript + Tailwind dashboard featuring real-time Recharts sparklines.

PUBLICATIONS

Bhosale, B., Shah, H., et al. (2025). *Profiling of PD-L1 and HER2 overexpression on cancer cells using AI-based macro-driven automation.* ESMO Open, 10(3), 105588. DOI: 10.1016/j.esmoop.2025.105588

ACHIEVEMENTS

- Finalist, NABARD Rural Innovation Hackathon (2024).